



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)
FULL SYLLABUS
23-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2021

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PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

- An object is projected upwards at an angle of 37° with horizontal with velocity 25 m/sec. Then find the position vector (in m) of particle after time 2 sec. ($g = 10 \text{ m/sec}^2$)
 (A) $40\hat{i} + 30\hat{j}$ (B) $40\hat{i} + 10\hat{j}$
 (C) $40\hat{i} - 10\hat{j}$ (D) $10\hat{i} + 40\hat{j}$
- Figure shows a hemisphere & a supported rod. Hemisphere is moving towards right with a uniform velocity v_2 & the end of the rod which is in contact with ground is moving in left direction with a velocity v_1 . Find the magnitude of rate at which the angle θ is changing in terms of v_1 , v_2 , R & θ .



- (A) $\frac{(v_1 + v_2) \sin^2 \theta}{R \cos \theta}$ (B) $\frac{(v_1 - v_2) \sin^2 \theta}{R \cos \theta}$
 (C) $\frac{(v_1 + v_2) \cos^2 \theta}{R \sin \theta}$ (D) $\frac{(v_1 - v_2) \cos^2 \theta}{R \sin \theta}$

- A neutron moving at a speed v undergoes a head on elastic collision with a mass number A at rest. Find the ratio of kinetic energies of neutron after & before collision.

- (A) $\left(\frac{1+A}{1-A}\right)^2$ (B) $\left(\frac{1-A}{1+A}\right)$
 (C) $\frac{1+A}{1-A}$ (D) $\left(\frac{1-A}{1+A}\right)^2$

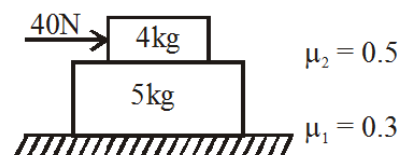
- Calculate the angle of banking required for a curve of 200 m radius so that a car rounding the curve at 80 kph would have no tendency to skid outward or inward. Assume the surface is friction less.

- (A) $\tan^{-1} \left(\frac{40}{81}\right)$ (B) $\tan^{-1} \left(\frac{30}{81}\right)$
 (C) $\tan^{-1} \left(\frac{20}{81}\right)$ (D) $\tan^{-1} \left(\frac{1}{4}\right)$

- Which of the following forces is not conservative?

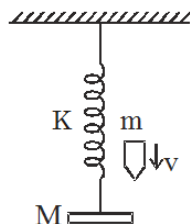
- (A) $\vec{F} = 3\hat{i} + 4\hat{j}$ (B) $\vec{F} = 3x\hat{i} + 4y\hat{j}$
 (C) $\vec{F} = 3y\hat{i} + 4x\hat{j}$ (D) $\vec{F} = 3x^2\hat{i} + 4y^2\hat{j}$

- Find the acceleration of the two blocks of 4 kg & 5 kg mass if a force of 40 N is applied on 4 kg block. Friction coefficient between respective surfaces are shown in fig.



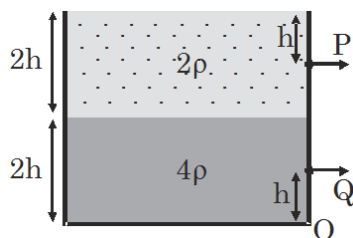
- (A) $5 \text{ m/sec}^2, 5 \text{ m/sec}^2$
 (B) $5 \text{ m/sec}^2, 2 \text{ m/sec}^2$
 (C) $2 \text{ m/sec}^2, 5 \text{ m/sec}^2$
 (D) $5 \text{ m/sec}^2, 0 \text{ m/sec}^2$

7. For the given system initially at rest as shown in figure, a bullet strikes the block inelastically and sticks to it then the amplitude of the system will be – (Neglect the thickness of the block)



- (A) $\frac{m}{k} \sqrt{\frac{v_0^2}{M+m} + g^2}$
 (B) $\frac{m}{k} \sqrt{\frac{v_0^2 k}{M+m} + g^2}$
 (C) $\frac{2m}{k} \sqrt{\frac{v_0^2}{M+m} + g^2}$
 (D) $\frac{m}{2k} \sqrt{\frac{v_0^2 k}{M+m} + g^2}$

8. A large tank is filled with two liquids of specific gravities 2ρ and 4ρ . Two holes are made on the wall of the tank as shown. Find the ratio of the distance from O of the points on the ground where the jets from holes P & Q strike.

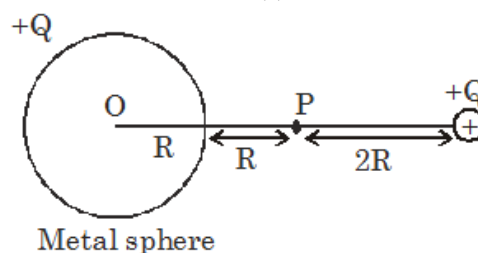


- (A) $\sqrt{3} : \sqrt{4}$ (B) $\sqrt{3} : \sqrt{2}$
 (C) $\sqrt{3} : 2$ (D) None of these

9. The displacement of a particle varies with time as $x = (15 \sin \omega t - 20 \sin^3 \omega t)$ cm.
 (a) Its motion is SHM
 (b) Its motion is not SHM
 (c) Maximum acceleration is $45\omega^2$
 (d) Time period of its motion is $(2\pi/\omega)$

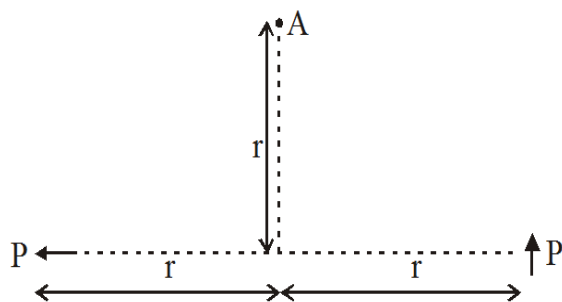
Choose the correct option :

- (A) (a) & (c) (B) (b) & (d)
 (C) (a), (c) & (d) (D) (b), (c) & (d)
10. A metal sphere is given charge $+Q$ as shown in the figure. A point charge $+Q$ is placed at a distance $4R$ from the centre of the sphere choose correct statement(s).

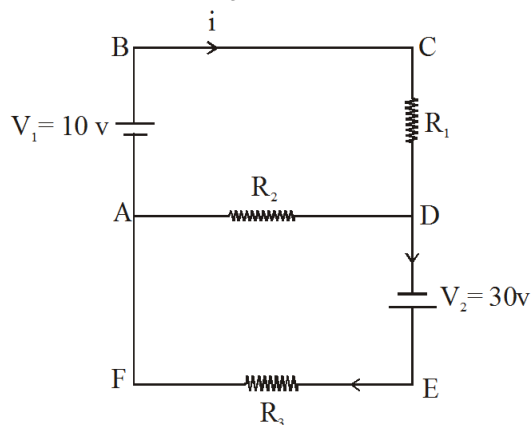


- (i) Net electric field at point P is zero
 (ii) In absence of point charge field at O is non zero.
 (iii) In presence of point charge field at P is towards metal sphere
 (iv) In presence of point charge field at O is zero
- (A) i, iii, iv are correct
 (B) i, ii are correct
 (C) ii, iii are correct
 (D) iii, iv are correct

11. P is the dipole moment of two small dipoles shown in the figure. What is the potential at point A?



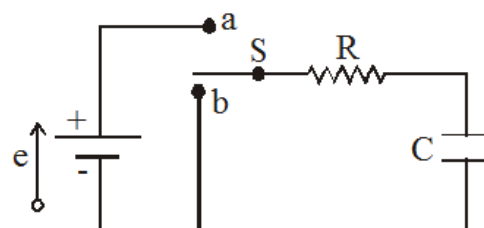
- (A) 0
(B) $\frac{Kp}{\sqrt{2}r^2}$
(C) $\sqrt{2} \frac{Kp}{r^2}$
(D) $2 \frac{Kp}{r^2}$
12. Two ideal batteries of emf V_1 and V_2 and three resistances R_1 , R_2 and R_3 are connected as shown in the below figure. The current in resistance R_2 is zero. If colour code of R_1 is red, green, yellow then colour code of R_3 will be.



- (A) Gray, Red, Violet
(B) Red, Green, Yellow
(C) Violet, Green, Yellow
(D) Green, Violet, Yellow

13. The table gives four sets of values for the circuit elements in figure. Rank the sets according to (as the switch is closed on point 'a' initially for long time) the time required for the current to decrease to half of its initial value, when switch is shifted at b, greatest first. Take time t_1 , t_2 , t_3 and t_4 respectively for four sets 1, 2, 3 & 4

	1	2	3	4
emf(V)	12	12	10	10
$R(\Omega)$	2	3	10	5
$C(\mu F)$	3	2	0.5	2



- (A) $t_4 > t_2 > t_1 > t_3$ (B) $t_4 > t_1 = t_2 > t_3$
(C) $t_3 > t_2 > t_1 > t_4$ (D) $t_3 > t_2 = t_1 > t_4$
14. Choose the correct statement
- (A) In most of northern hemisphere the northpole of the dip needle tilt upwards
(B) In most of northern hemisphere the northpole of the dip needle tilt downwards
(C) In most of southern hemisphere the northpole of the dip needle tilt downwards
(D) At equator the northpole of the dip needle tilts downwards.

15. A domain in ferromagnetic iron is in the form of a cube of side length $1 \mu\text{m}$. The molecular mass of iron is 56 g/mole and its density is 8 g/cm^3 . Assume that each iron atom has a dipole moment of $9.1 \times 10^{-23} \text{ Am}^2$. Take Avogadro number 6×10^{23} .

- (i) Number of atoms in domain $= 8.2 \times 10^{12}$
 (ii) Maximum possible dipole moment of the domain $= 7.8 \times 10^{-12} \text{ Am}^2$
 (iii) Maximum magnetisation of domain is $= 7.8 \times 10^6 \text{ A/m}$.

- (A) all are correct
 (B) only (i) & (ii) are correct
 (C) only (ii) & (iii) are correct
 (D) only (i) & (iii) are correct

16. In an AC circuit, $C = \frac{25}{\pi} \mu\text{F}$ and $R = 300\Omega$ are connected in series with AC source of 200V and 50 sec^{-1} frequency. The power dissipated and power factor in circuit will be :

- (A) 48 W , 0.4 (B) 24 W , 0.6
 (C) 48 W , 0.6 (D) 24 W , 0.4

17. Which statement regarding work function (ϕ) is correct?

- (A) Work function depends on intensity of incident radiation
 (B) Work function depends on frequency of incident radiation
 (C) Work function depends on area of metal surface
 (D) Work function depends on the surface of the metal.

18. Which of the following band gap semiconductor is suitable for LED?

- (A) 1.2 eV (B) 2 eV
 (C) 4 eV (D) Any of the above

19. Transistors are :

- (A) Current controlling devices
 (B) Voltage controlling device
 (C) Resistance controlled device
 (D) none of these

20. In an amplitude modulated wave for audio-frequency of 500 Hz , the appropriate carrier wave frequency will be :-

- (A) 50 Hz (B) 100 Hz
 (C) 500 Hz (D) $50,000 \text{ Hz}$

SECTION-II : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

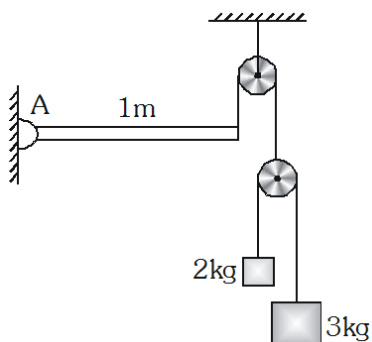
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25 , 7.00 , -0.33 , $-.30$, 30.27 , -127.30 , if answer is $11.36777\dots$ then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

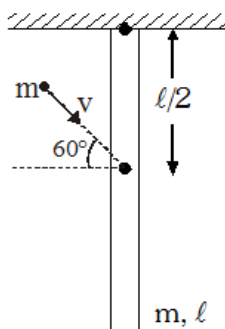
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

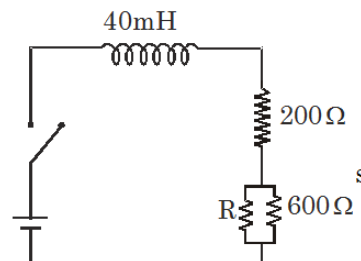
1. Consider the situation shown in the figure. Uniform rod of length $L = 1\text{m}$ can rotate freely about the hinge A in vertical plane. Pulleys and strings are light and frictionless. If the rod remains horizontal at rest during the motion of masses, then the mass of the rod (in kg) is



2. A thin rod of mass m and length ℓ is hinged to a ceiling and it is free to rotate in a vertical plane. A particle of mass m , moving with speed v strikes it as shown in the figure and gets stick with the rod. The value of v (in m/s), for which the rod becomes horizontal after collision is (given $\ell = 3\text{m}$, $m = 1\text{kg}$ & $g = 10\text{ m/s}^2$ and $\sqrt{35} = 5.9$)



3. Two capillary tubes of same diameter are put vertically one each in two liquids whose relative densities are 0.4 and 0.3 and surface tensions are 50 dyne/cm and 25 dyne/cm respectively. Ratio of heights of liquids in the two tubes $\frac{h_1}{h_2}$ is :-
(Take angle at contact = 0°)
4. When switch S is closed, then for what value of R (in Ω) will the time constant will be 100 μs .



5. In an LC oscillator, $L = 12\text{ mH}$ & $C = 1.6\text{ }\mu\text{F}$ and $R = 1.5\text{ }\Omega$. Find the time (in ms) in which max charge on capacitor falls to half of its initial value (approximately)
6. Suppose the energy of k_α x-rays of two elements P & Q are 826.2 eV and 2947.8 eV then the number of elements lying between P & Q according to their atomic number is
7. Angular magnification for an astronomical telescope is 5 and the tube length for image at infinity is 36cm. Then find tube length (in cm) for image formed at least distance of distinct vision for normal eye (25 cm)
8. A light of wavelength 5000 $\text{\AA}$ falls on a liquid film of refractive index $\frac{4}{3}$ making angle of refraction 60° . The width of film (in 10^{-7}m) so as the reflected rays appear dark is
9. A parallel beam of monochromatic light of wavelength 450 nm passes through a long slit of width 0.2 mm. Find the angular divergence (in 10^{-3} rad) in which most of the light is diffracted :-
10. The value of one division of a vernier scale in a device is $(0.4)^\circ$ and 20 divisions of main scale are equal to 25 divisions of vernier. Then leastcount of device (in degree) is :

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.



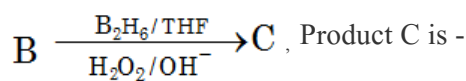
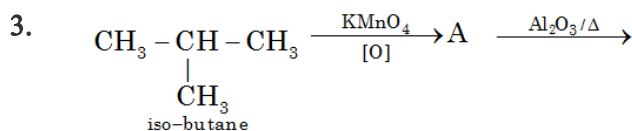
Backward reaction is endothermic. If some quantity of Cl_2 is added keeping volume of container constant which of the following statement is correct.

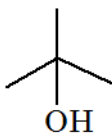
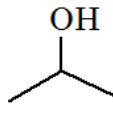
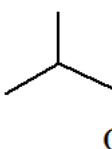
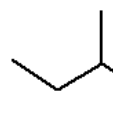
- (A) No effect on instantaneous pressure of container
(B) Temp of container will increase
(C) Temp of container will decrease
(D) No effect on temperature of container

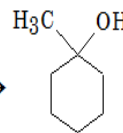
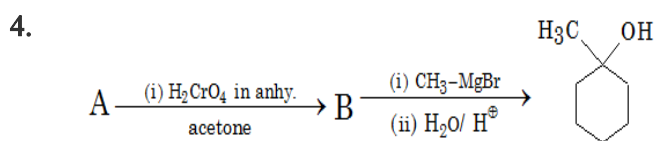
2. What will be the total number of ammonia molecules required to completely cover surface area of 20 gm charcoal by monolayer formation. Given that 1 gm charcoal has surface area 10^3 m^2 and radius of ammonia molecule is $r \text{ nm}$.

Given that : ammonia molecules form hexagonal close packing in two dimension)

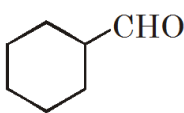
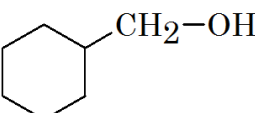
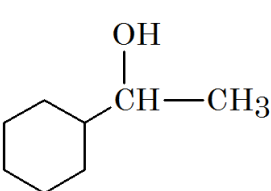
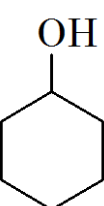
- (A) $\frac{2 \times 10^{22}}{\pi r^2}$ (B) $\frac{10^{22}}{\sqrt{3}r^2}$
(C) $\frac{2 \times 10^3}{\pi r^2}$ (D) $\frac{4 \times 10^{18}}{\pi r^2}$



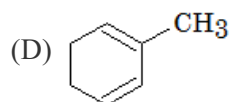
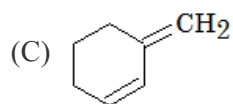
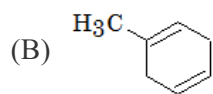
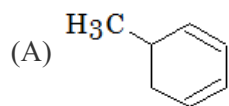
- (A)  (B) 
(C)  (D) 



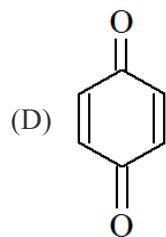
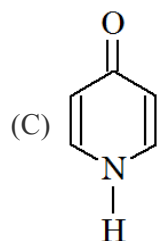
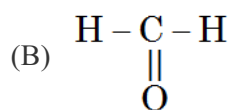
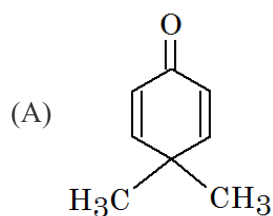
identify structure of "A"

- (A)  (B) 
(C)  (D) 

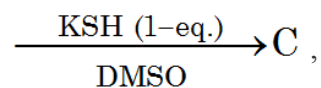
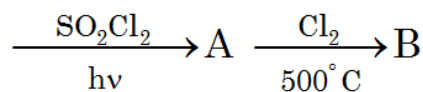
5. Which of the following alkene is most stable ?



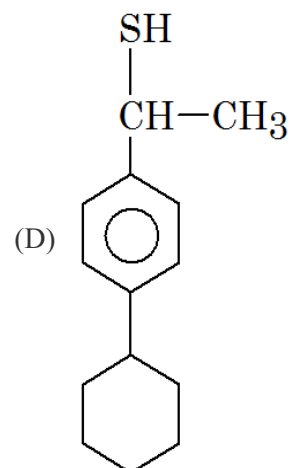
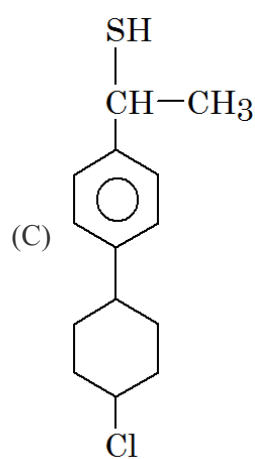
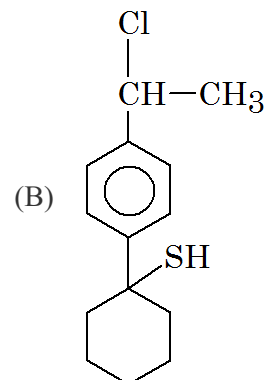
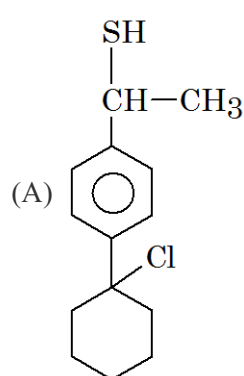
6. Which of the following compound show tautomerism



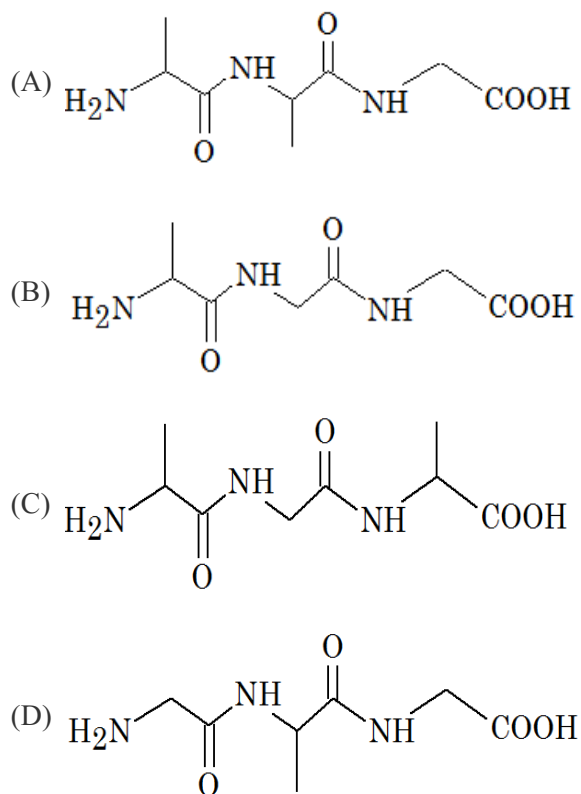
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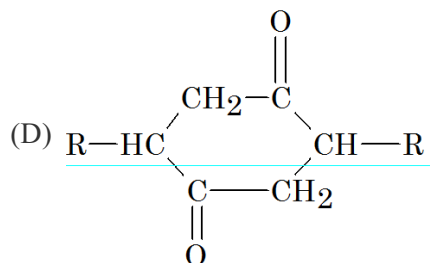
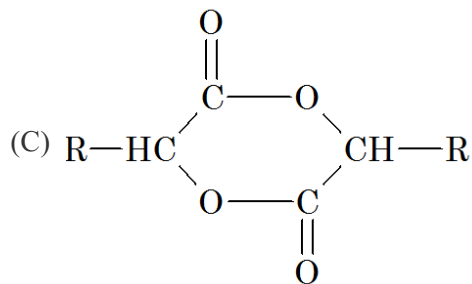
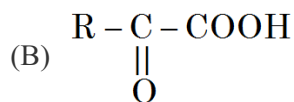
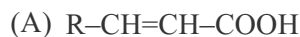
Product C is :



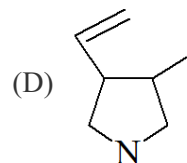
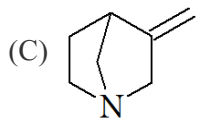
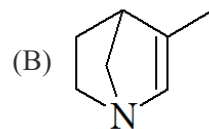
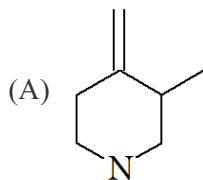
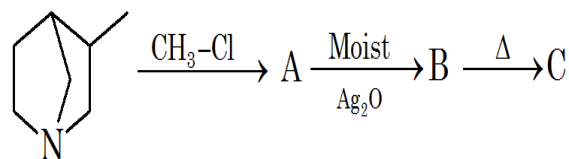
8. The correct structure of tripeptide made up of Alanine-Glycine-Alanine is :



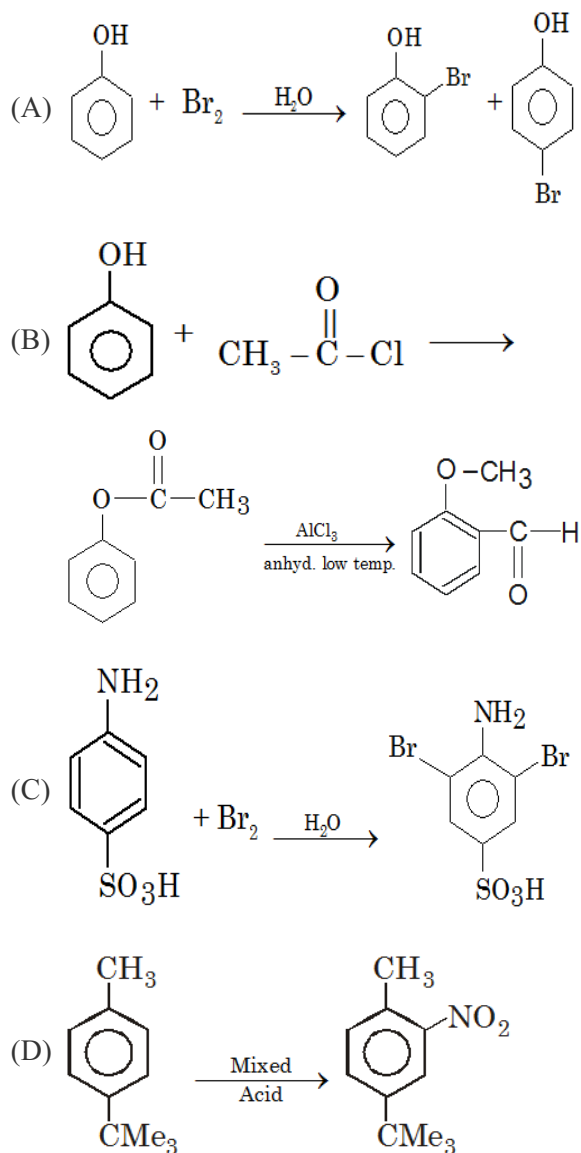
9. Compound RCH_2CO_2H (A) on reaction with Br_2 /Red P gives B which on reaction with aq KOH gives C which on heating gives D. Find out D.



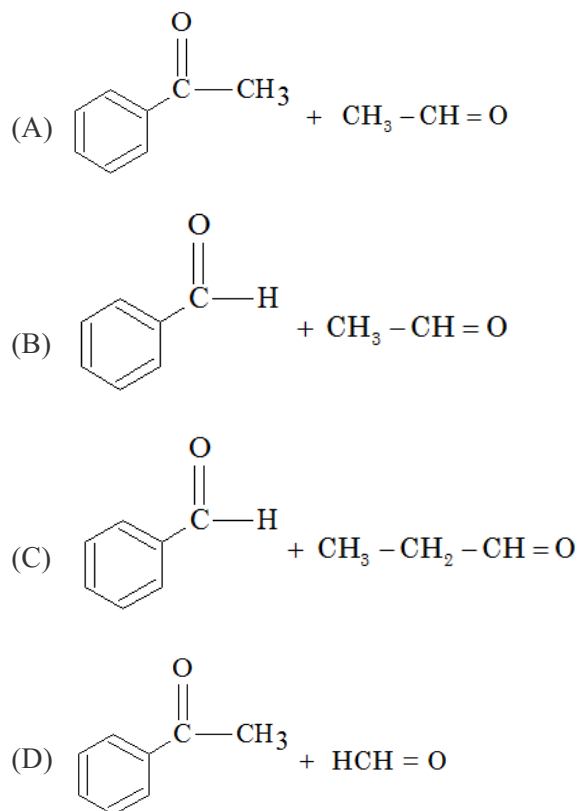
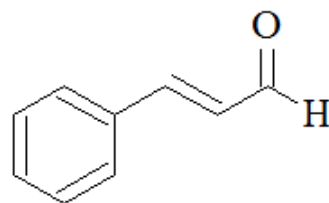
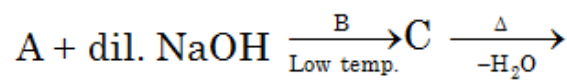
10.



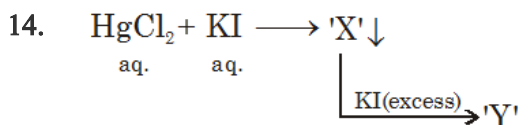
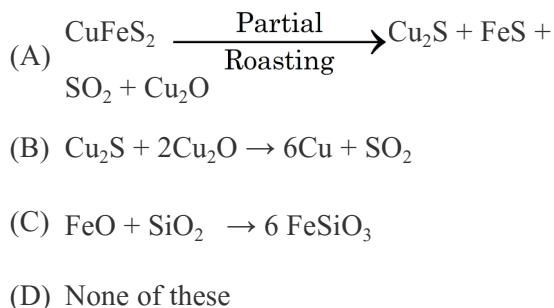
11. Out of the following which is correct :



12.



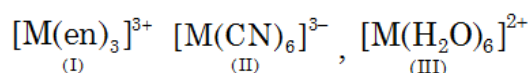
13. During the extraction of Cu from CuFeS_2 which of the following reaction does not occur :-



Select correct option :-

- (A) 'Y' is a mixture of black ppt. and water soluble complex
- (B) 'X' is green ppt.
- (C) 'Y' have sp^3 hybridisation of Hg
- (D) Neutral solution of 'Y' is used to detect NH_3 gas
15. Which of the following does not exist?
- (A) ClI_3 (B) XeH_4
- (C) BiF_5 (D) Both (A) and (B)

16. For three complex compounds



correct information is :-

- (A) Order of Δ_0 will be $\text{I} > \text{II} > \text{III}$
- (B) Order of λ_{abs} will be $\text{III} > \text{II} > \text{I}$
- (C) All complex will show optical isomerism
- (D) All complex will not show geometrical isomerism

17. Which one has highest 2nd I.P value?

(A) Mn (B) Cr (C) V (D) Ti

18. Which is the correct maximum prescribed concentration of metals in ppm in drinking water

(A) Al = 0.2 (B) Fe = 0.3

(C) Zn = 10 (D) Cu = 6

19. The metal chloride which produces apple green coloured flame when directly subjected to flame.

(A) Barium (B) Strontium

(C) Calcium (D) Sodium

20. What are the names assigned to elements after uranium

(A) Transition elements

(B) Inner transition elements

(C) Trans uranic elements

(D) Transfermium elements

SECTION-II : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

- One mole of gas 'A' and 2 moles of SO_2 are placed in a container, gaseous mixture effusing out initially from a pinhole made in container has ratio of A & SO_2 in 1 : 1 mole ratio. The molar mass of gas A is :
- If uncertainty in position of electron is equal to half of its de Broglie wave length. The minimum percentage error in its measurement of velocity under this condition, will be approximately (Nearest integer)
- Assuming that bond enthalpy of axial P-Cl bond is different from bond enthalpy of equatorial P - Cl bond, bond enthalpy of axial P - Cl bond in PCl_5 will be (in KJ/mol) :
(given: $\Delta H_f \text{PCl}_3(\text{g}) = -320 \text{ KJ/mol}$)
 $\Delta H_f \text{P}(\text{g}) = 315 \text{ KJ/mol}$
 $\Delta H_f \text{PCl}_5(\text{g}) = -400 \text{ KJ/mol}$
 $\Delta H_{\text{atomization}} \text{Cl}_2 = 240 \text{ KJ/mol}$
& bonds of PCl_3 are same as equatorial bonds of PCl_5
- On doubling the temperature of reaction from initial temperature of 200 K, fraction of activated molecule become $\sqrt{e}$ times of its initial value, activation energy of reaction is (in calories) - [Given : $R \approx 2 \text{ cal K}^{-1} - \text{mol}^{-1}$]
- Aqueous solution of NaCl is electrolysed by using a variable current source where current as expressed as $i = (2 + t)$. If electrolysis is carried out for 20 second then find milimole of H_2 evolved. (i = current in ampere and t = time in seconds)
- Find the number of incorrect statements from following list -
(1) Enthalpy of formation of a compound always have positive value.
(2) Standard enthalpy of formation of $\text{OH}^-_{(\text{aq})} = 0$
(3) Enthalpy of formation of carbon mono oxide gas at 300 K is -110 kcal/mol then internal energy change for same process will be -109.7 kcal/mol
(4) Standard enthalpy of formation of $\text{H}^+_{\text{aq}} = 0$
- Emf of given cell
 $\text{Zn}_{(\text{s})} | \text{Zn}^{++}_{\text{aq}} (0.1\text{M}) || \text{Fe}^{++}_{\text{aq}} (0.01\text{M}) | \text{Fe}_{(\text{s})}$
is 0.3 volt. The value of $\log K_{\text{eq}}$ of cell reaction will be—
(Take : $\frac{2.303RT}{F} = 0.06$)
- 100 mL, 0.1 M $\text{CH}_3\text{COONH}_4$ solution has pH x_1 . Now by keeping temperature constant 400mL of water was added to solution and on calculation pH was found to be x_2 . The ratio of x_1 to x_2 will be —
(Given : $\text{pK}_a(\text{NH}_4^+) = 9.3$; $\text{pK}_a(\text{CH}_3\text{COOH}) = 4.7$)
- Octahedral complex $\text{Ma}_x\text{b}_y\text{c}_z$ have two geometrical isomers, both are optically inactive then correct value of $(x \times y \times z)$ is :-
- What is the bond order of C_2

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Let $f_1 = \sum_{i=1}^3 a_i x_i$, $f_2 = \sum_{i=1}^3 b_i x_i$ and $f_3 = \sum_{i=1}^3 c_i x_i$. If the equations $f_1 = f_2 = f_3 = 0$ has a

non-trivial solution, then

(A) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$

(B) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$

(C) $\begin{vmatrix} a_1 & b_1 & c_3 \\ a_2 & b_2 & c_2 \\ a_1 & b_3 & c_3 \end{vmatrix} = 0$

(D) $\begin{vmatrix} a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \\ a_3 & b_3 & 1 \end{vmatrix} = 0$

2. If p : It rains today, q : I go to school, r : I shall meet any friends and s : I shall go for a movie, then which of the following is the proposition :
If it does not rain or if I do not go to school, then I shall meet my friend and go for a movie.

(A) $\sim(p \wedge q) \Rightarrow (r \wedge s)$

(B) $\sim(p \wedge \sim q) \Rightarrow (r \wedge s)$

(C) $\sim(p \wedge q) \Rightarrow (r \vee s)$

(D) none of these

3. Probability that a student is passed in mathematics is $1/4$ and probability that he is passed in physics is $1/5$. Probability that he is passed in both the subjects is $3/20$. What is the probability that he is passed in mathematics if he has been failed in physics.

(A) $3/4$ (B) $1/8$

(C) $3/16$ (D) $5/8$

4. Let z_1, z_2 are any complex numbers which satisfies the following conditions $|z_1| = 12$ and $|z_2 - \sqrt{3} - i| = 5$ respectively then range of values of $|z_1 - z_2|$ will be

(A) $[6, 18]$ (B) $[5, 19]$

(C) $[5, 13]$ (D) $[6, 19]$

5. Let $A(h, k)$, $B(1, 1)$ and $C(2, 1)$ be vertices of a right angled triangle with AC as its hypotenuse. If area of triangle is 2, then value of $h^2 + k^2$ can be

(A) 25 (B) 9

(C) 26 (D) 11

6. Let $3x + 4y + z - 3 = 0$ ($P_1 = 0$) and $6x + 8y + 2z - 5 = 0$ ($P_2 = 0$) be the two given planes, then equation of plane which divides the distance between the planes $P_1 = 0$ and $P_2 = 0$ in the ratio 2 : 3 is
- (A) $3x + 4y + z + 14 = 0$
 (B) $15x + 20y + 5z - 14 = 0$
 (C) $3x + 4y + z - 14 = 0$
 (D) $15x + 20y + 5z + 14 = 0$
7. The greatest value of $f(x) = \log_{\sqrt{48}}(x^2 - 4x + 3)$ on interval $[-5, 5]$ is
- (A) 1 (B) $\frac{1}{2}$ (C) 2 (D) 4
8. $\tan^{-1} \frac{3}{5} + \tan^{-1} \frac{3}{29} + \tan^{-1} \frac{3}{71} + \tan^{-1} \frac{3}{131} + \dots$ (n terms) =
- (A) $\tan^{-1} \frac{3}{2+3n}$ (B) $\tan^{-1} \frac{3n}{2n+3}$
 (C) $\tan^{-1} \frac{3n}{1+4n}$ (D) $\tan^{-1} \frac{3n}{2+3n}$
9. Area bounded by curve $y^2 = 4x$ and lines $y = x$ and $y = 2x$ is
- (A) 1 (B) $\frac{7}{3}$ (C) $\frac{8}{3}$ (D) $\frac{1}{3}$
10. A vertical lamp post of height 8m stands at the centre of a equilateral triangular field. The angle of elevation of its top from each vertices of triangle is 45° . The area of triangular field is :
- (A) $16\sqrt{3}$ (B) $32\sqrt{3}$
 (C) $48\sqrt{3}$ (D) $72\sqrt{3}$
11. If shortest distance between the skew lines $r = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ and $r = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k})$ is ℓ , then value of $[\ell]$ is : (where $[\]$ is greatest integer function)
- (A) 1 (B) 2 (C) 3 (D) 4
12. If A is a square matrix of order n such that $|\text{adj}(\text{adj}A)| = |A|^{P^2}$, (where P is a prime number), then n can not be :
- (A) 4 (B) 8 (C) 10 (D) 12
13. If 9^{th} , 13^{th} and 15^{th} terms of an A.P. are first three terms of a G.P. whose sum of infinite terms is 80, then first term of A.P. is :
- (A) 40 (B) 64 (C) 80 (D) 96
14. If a variable tangent of the circle $x^2 + y^2 = 1$ intersect the ellipse $x^2 + 2y^2 = 4$ at P and Q, then locus of point of intersection of tangents at P and Q is an ellipse whose eccentricity is equal to :
- (A) $\sin 36^\circ$ (B) $\cos 36^\circ$
 (C) $\cos 60^\circ$ (D) $\sin 60^\circ$
15. The value of ' μ ' for which the line $y = \mu x + 2$ become a tangent to the hyperbola $4x^2 - 9y^2 = 36$ is
- (A) $\pm \frac{2}{3}$ (B) $\pm \frac{2\sqrt{2}}{3}$
 (C) $\pm \frac{8}{9}$ (D) $\pm \frac{4\sqrt{2}}{3}$

16. If the tangent to the curve $y^2 = x^3$ at the point (α^2, α^3) is also normal to the curve at the point (β^2, β^3) , then the value of $\alpha\beta$ is

(A) $-\frac{1}{9}$ (B) $-\frac{2}{9}$
(C) $-\frac{1}{3}$ (D) $-\frac{4}{9}$

17. If x and y are positive integers, then ${}^xC_r + {}^xC_{r-1} {}^yC_1 + {}^xC_{r-2} {}^yC_2 + \dots + {}^yC_r$ is equal to (wherever defined)

(A) $\frac{x!y!}{r!}$ (B) $\frac{(x+y)!}{r!}$
(C) ${}^{x+y}C_r$ (D) ${}^{xy}C_r$

18. The mean and variance of a random variable x having a binomial distribution are 4 and 2 respectively, then the value of $P(x=1)$ is

(A) $\frac{1}{4}$ (B) $\frac{1}{16}$
(C) $\frac{1}{8}$ (D) $\frac{1}{32}$

19. How many three digit numbers satisfy the property that the middle digit is arithmetic mean of the first and the last digit

(A) 41 (B) 45 (C) 43 (D) 44

20. A parabola touch lines $x - y = 0$ and $x + y + 2 = 0$ at points $M(1, 1)$ and $N(0, -2)$ respectively, then the equation of the directrix is :

(A) $3x + y + 4 = 0$ (B) $x - 3y - 2 = 0$
(C) $8x - y + 7 = 0$ (D) None of these

SECTION-II : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. Let $f(x) = \sin 2x$ and $g(x) = \ln x$, then total number of solutions of $f(|x|) = |g(x)|$ will be ($x \in (0, \pi)$)

2. If $y(x)$ is solution of $x \frac{dy}{dx} + 3y = x$, $y(2) = 1$ then value of $y(1)$ is

3. If $A_n = \int_0^{\pi/2} \frac{\sin(2n-1)x}{\sin x} dx \quad \forall n \in \mathbb{N}$ then value of $\frac{A_{1000}}{A_{10}}$ is

4. $\lim_{x \rightarrow 0} \frac{\cos 5x \cos 2x - \cos 7x \cos 10x}{2 \ln^2(\sin x + \cos x)^2}$ is equal to :

5. If $\vec{x} = \hat{i} - \hat{j}$, $\vec{y} = \hat{i} + \hat{j} + \hat{k}$ are two vectors and $\vec{z}$ is another vector such that $\vec{x} \times \vec{z} + \vec{y} = \vec{0}$ and $\vec{x} \cdot \vec{z} = 0$, then $|\vec{z}|^2$ is equal to

6. Let $f(x) = ax + b$ is a function such that $f^{-1}(2) = 3$, $f^{-1}(-3) = 6$, (where $f^{-1}(x)$ is the inverse of function $f(x)$) then $a + b$ is equal to
7. Each of "n" urns contain 4 white and 6 black balls. The $(n + 1)^{\text{th}}$ urn contain 5 white and 5 black balls. One of the urn is chosen at random and two balls are drawn from it without replacement. Both the balls turn out to be black. If the probability that the $(n + 1)^{\text{th}}$ urn was chosen to draw the ball is $\frac{1}{16}$, then value of $\frac{n}{4}$ is
8. If $2f(x) + xf\left(\frac{1}{x}\right) - 2f\left\{\left|\sqrt{2}\sin\pi\left(x + \frac{1}{4}\right)\right|\right\} = 4\cos^2\frac{\pi x}{2} + x\cos\frac{\pi}{x} \forall x \in \mathbb{R} - \{0\}$, then $\left(f(2) + f(1) + 3f\left(\frac{1}{2}\right)\right)$ equals
9. If $\cos^4\theta + \alpha$, $\sin^4\theta + \alpha$ are the roots of equation $x^2 + 4x + 2 = 0$ and $\cos^2\theta + \beta$, $\sin^2\theta + \beta$ are the roots of equation $x^2 + 2bx + b = 0$, then sum of all possible values of b is equal to (Where $\alpha, \beta \in \mathbb{R}$)
10. If $g(x)$ is non-negative continuous function such that $g(x) + g\left(x + \frac{1}{2}\right) = 2$ then the value of $\int_0^1 (\sin(8\pi x)) g(x) dx =$

Space for Rough Work